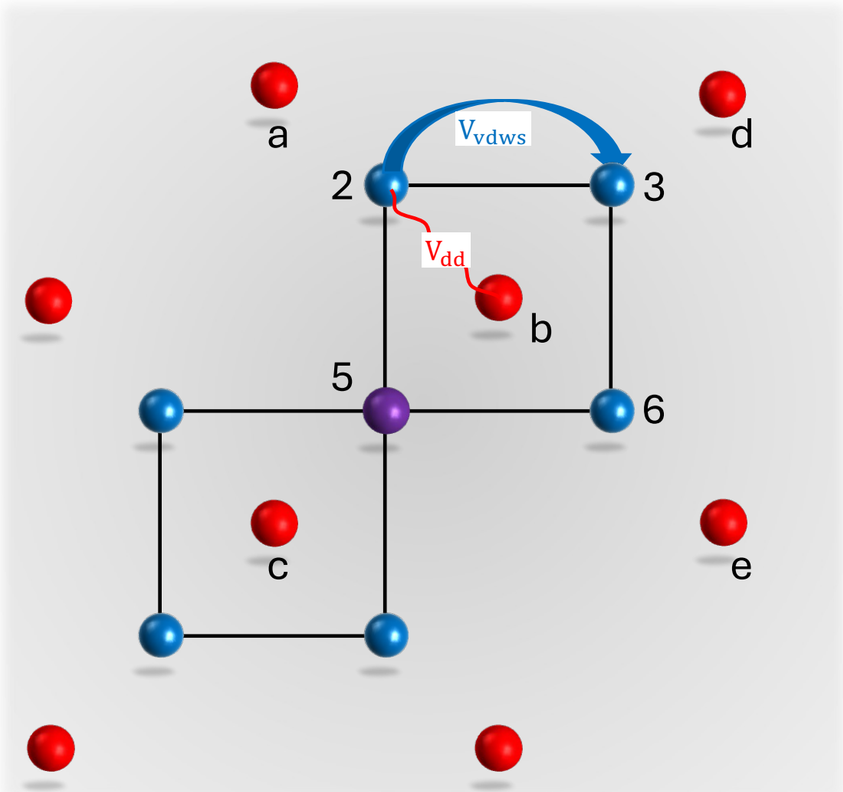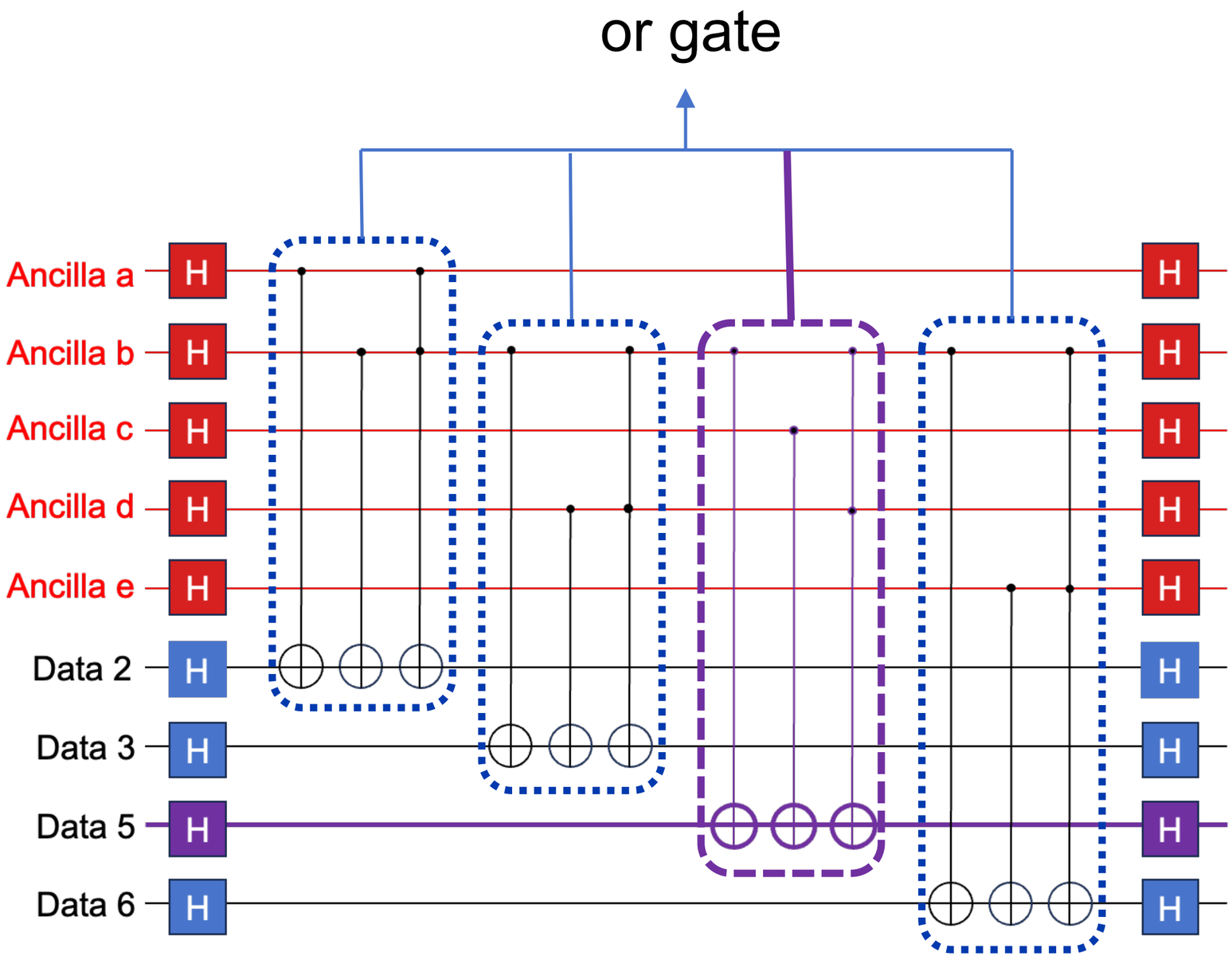X-error correction model